

Polynomial approximation pseudo-two-dimensional battery model for online application in embedded battery management system



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ABSTRACT

Physics-based model has been regarded as a promising alternative to equivalent circuit model due to its ability to describe internal electrochemical states of battery. However, the rigorous physics-based model, namely pseudo-two-dimensional (P2D) model, is too complicated for online application in embedded battery management system. In this paper, to simplify the P2D model, a series of polynomial functions are employed to approximate the electrolyte phase concentration profile, solid phase concentration profile, and non-uniform reaction flux profile, respectively. Especially, the accuracy of 2nd-order and 3rd-order polynomial approximations for reaction flux is compared, and the higher-order is validated with more strength. Benefit from the acquisition of above variables, the electrolyte potential is derived directly according to the conservation of charge at electrolyte phase; the accuracy of activation overpotential is also improved by using the non-uniform reaction flux rather than assuming the uniform current density in single particle (SP) model. Finally, the developed model is simulated by different constant current rates, hybrid pulse and driving cycles, and its outputs are compared with P2D model and original SP model. The results demonstrate that the model proposed in this paper could capture the battery characteristics efficiently, and also significantly reduce the computation complexity.

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1. Introduction

Nowadays, under the pressure of fossil fuel depletion and emission pollution, many governments of different regions and countries among the world have turned to develop the electric vehicles (EVs) and hybrid electric vehicles (HEVs). With the superiority of high specific energy and power, long life, low self-discharge rate, non-memory effect, lithium-ion batteries have been regarded as the preferred candidate for these vehicles [1]. To guarantee safe and reliable battery operations, and prolong the battery life, a battery management system (BMS) is indispensable to online monitor the battery states and regulate the charge or discharge behaviors.

Since most mature and sophisticated algorithms for state estimation are model-based, it is prerequisite to construct a battery model with high fidelity and easy to application. Two

categories battery models have been proposed and researched in the literature, namely empirical equivalent circuit models (ECMs) and physics-based electrochemical models [2,3]. Although comprised of circuit components, the ECMs can efficiently model the external characteristics (current-voltage relationship) of battery. And by applying the data-driven algorithms [4], the model parameters of ECMs can be online identified to adapt the operation condition. However, the parameters of ECMs cannot embody electrochemical meanings and ensure high model accuracy at whole operation ranges, especially at low state of charge (SOC) or low temperature [5]. The rigorous physics-based battery model originally developed by Doyle, Fuller and Newman [6,7] is a pseudo-two-dimensional (P2D) porous electrode model, which consists of four coupled partial differential equations (PDEs) and one algebraic equation. Compared with the ECMs, this physics-based model can not only predict the external

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characteristics of battery, but also give insight into the internal electrochemical states of battery. The advantages of knowing these internal electrochemical states include analyzing degradation mechanism [8], extending battery life, predicting power limits [9], designing optimal charge strategies [10] and so on. However, the coupled PDEs drastically increase the computing burden of running the model and make it impractical to be applied into an embedded controller.

To simplify the rigorous P2D model, plenty of methods have been developed with different degrees of fidelity and model complexity. From the orientation of control, Smith et al. [11] and Lee et al. [12] derive the analytical transfer functions of electrochemical variables by solving the governing PDEs. In further study, Lee et al. [13,14] employ a discrete-time realization algorithm to convert the transcendental transfer functions into discrete-time state-space models, and propose a model-blending method to extend the operation from a fixed equilibrium point over a wide range of SOC and temperatures. In our previous study [15], a simple and efficient way based on SOC change is also proposed to extend the model accuracy over a wide SOC range. Besides, Padé approximation has been widely applied to convert the transcendental transfer functions into lower-order and fractional transfer functions, especially for solid phase diffusion [16]. The drawback of using linear transfer functions is that the model error could increase gradually as the operation point deviates from the fixed equilibrium point.

Since most computing burden of electrochemical model come from the solid phase diffusion in the additional pseudo dimension, some researchers focus on simplifying the solid phase diffusion equation. Subramanian et al. [17] assume the lithium ion (Li^+) concentration in each particle follow the parabolic profile (PP), and use the volume-averaged method to compute the polynomial coefficients. And a three-parameter polynomial profile model which has been validated in the literature could approximate the solid phase concentration distribution efficiently. In the PP model, all other equations remain the same as the P2D model. While the most simplified model is single particle (SP) model developed by Haran et al. [18] for metal hydride battery and Zhang et al. [19] for lithium ion battery. The SP model represents each electrode by a single spherical particle, and completely neglects Li^+ concentration and potential inside the electrolyte phase. With these assumptions, the SP model presents lower computing burden and high accuracy at lower C-rates, but significant errors happen at higher C-rates. Santhanagopalan et al. [20] compare the SP and PP models with the P2D model, and prove that the PP model significantly reduces the computational time, while retains sufficient accuracy; the large error of the SP model at higher C-rates mainly comes from the ignored electrolyte dynamics. Then different methods are proposed to enhance the accuracy of SP model by taking the effect of electrolyte dynamics into consideration. Laplace transform with separation of variables (SOVs) method [21], and integral method analysis (IMA) [22] are employed to derive the analytical transfer functions of electrolyte concentration. Li et al. [23] propose a first-order process to model the electrolyte concentration polarization, and the related parameters are identified by least square fit method. Polynomial functions are applied to approximate the electrolyte concentration and potential profiles by Rahimian et al. [24]. However, the above methods are all based on the assumption of uniform reaction flux, which is clearly impossible and could lead to relatively large errors. For constant current input,

the steady state of electrolyte concentration distribution produces a parabolic profile in each electrode and a straight line in separator. Luo et al. [25] propose a time-varying coefficient to consider the transition process from the initial uniform distribution to the parabolic one. By directly assuming the parabolic distribution of electrolyte concentration, Han et al. [26] derive the linear equations to calculate the coefficients of polynomial functions without presupposing the uniform reaction flux. After the electrolyte concentration is acquired, the electrolyte potential difference between the two current collectors can be calculated according to the simplified formula developed by Prada [27] based on the assumption of uniform reaction flux.

Without considering the contact resistance, the terminal voltage of battery consists of four parts: the equilibrium potential, the electrolyte potential, the activation overpotential, and the overpotential caused by the solid electrolyte interface (SEI) film resistance. All of them are directly determined by the reaction flux at the two current collector interfaces. Since the reaction flux is non-uniform in each electrode and even change over time under the galvanostatic discharge condition [28]. Using the volume-averaged reaction flux in the SP model to calculate the four parts could lead to enough errors. Taking use of the electrode and electrolyte potential equations, Butler-Volmer equation and potential balance relation, Luo et al. [28] and Han et al. [29] derive two different formulas to calculate the non-uniform reaction flux. However, both of them need to calculate the distribution of equilibrium potential along the x-axis, and the uniform current distribution has to be assumed to calculate intermediate variables.

In this paper, the polynomial approximation technology is extensively used to simplify the P2D model. The PP method is employed to solve the solid phase diffusion; the distribution of electrolyte concentration is imitated by the parabolic functions; and the non-uniform reaction flux is obtained by the approximation method proposed in this paper. Especially, the accuracy of 2nd-order and 3rd-order polynomial approximations for reaction flux is compared, and the higher-order is validated with more strength. Benefit from the acquisition of electrolyte concentration and non-uniform reaction flux, the electrolyte potential is derived directly according to the conservation of charge at electrolyte phase; the accuracy of activation overpotential is improved when compared with the SP model using uniform reaction flux. The obvious superiority of polynomial approximation is its lower computational burden, which makes the simplified model feasible to be applied into an embedded controller. The developed model is simulated by different discharge C-rates, hybrid pulse and urban dynamometer driving schedule (UDDS) cycles. And the results are compared with P2D model and original SP model to validate the improvement in computational efficiency and model accuracy, respectively.

The remaining contents of this paper are arranged as follows: Section 2 introduces the principle of pseudo-two-dimensional model; Section 3 illustrates the method to approximate the distribution of electrolyte concentration; Section 4 illustrates the method to approximate the distribution of local volumetric current density; the simplified model are constructed in Section 5; the simulation results, discussion and validation are presented in Section 6; Section 7 concludes this paper. To facilitate better understanding of the acronyms and variables used throughout this paper, the list of all acronyms and nomenclature are presented in the end part of this section.

Acronyms			
BMS	battery management system	PP	parabolic profile
cc-neg	negative current collector	P2D	pseudo-two-dimension
cc-pos	positive current collector	RMSE	root mean square error
ECMs	empirical equivalent circuit models	SEI	solid electrolyte interface
EVs	electric vehicles	SOC	state of charge
HEVs	hybrid electric vehicles	SOVs	separation of variables
IMA	integral method analysis	SP	single particle
Li^+	lithium ion	UDDS	urban dynamometer driving schedule
MAE	maximum error	2nd app	2nd-order polynomial approximation
PDEs	partial differential equations	3rd app	3rd-order polynomial approximation
PP2D	polynomial approximation pseudo-two-dimensional model		

Nomenclature			
A	electrode plate area, m^2	T	temperature, K
a_s	specific interfacial surface area, $m^2 m^{-3}$	t	time, s
c_e	electrolyte concentration, $mol m^{-3}$	t^+	lithium ion transference number
$c_{e,0}$	average electrolyte concentration, $mol m^{-3}$	U	electrode equilibrium potential, V
c_s	solid phase concentration, $mol m^{-3}$	V	terminal voltage of battery cell, V
$c_{s,e}$	solid phase surface concentration, $mol m^{-3}$	x	ID coordinate across the cell, m
$\bar{c}_s$	volume-averaged solid phase concentration, $mol m^{-3}$	z	ID coordinate across electrode/seperator, m
D_s	solid phase diffusivity, $m^2 s^{-1}$	ε_s	active material volume fraction
D_e	electrolyte phase diffusivity, $m^2 s^{-1}$	ε_e	electrolyte volume fraction
F	Faraday's constant, $96487 C mol^{-1}$	θ	stoichiometry of electrode concentration
I	load current, A	θ_0	stoichiometry at 0% SOC
i_0	exchange current density, $A m^{-2}$	θ_{100}	stoichiometry at 100% SOC
j	reaction flux, $mol m^{-2} s^{-1}$	α_a, α_c	charge transfer coefficient
j_f	local volumetric current density, $A m^{-3}$	σ	solid phase conductivity, $S m^{-1}$
k	reaction rate constant, $mol^{-0.5} m^{2.5} s^{-1}$	κ	electrolyte phase conductivity, $S m^{-1}$
L	thickness, m	φ_s	solid phase potential, V
$\bar{q}$	volume-averaged concentration flux, $mol m^{-3} m^{-1}$	φ_e	electrolyte phase potential, V
R	universal gas constant, $8.31451 J mol^{-1} K^{-1}$	η	activation overpotential, V
R_s	particle radius, m	$d \ln f_{\pm} / d \ln c_e$	activity dependence
R_{SEI}	SEI film resistance, Ωm^2	dt	sampling step, s
r	radial coordinate, m		

Table 1

The governing equations of the P2D model for lithium-ion battery.

Governing equations	Boundary conditions
Negative/Positive electrode $\frac{\partial c_s}{\partial t} = \frac{D_s}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right)$ $\varepsilon_e \frac{\partial c_e}{\partial t} = D_e^{\text{eff}} \frac{\partial^2 c_e}{\partial x^2} + (1 - t_+) \frac{j_f}{F}$ $\sigma^{\text{eff}} \frac{\partial^2 \varphi_s}{\partial x^2} = j_f$ $\kappa^{\text{eff}} \frac{\partial^2 \varphi_e}{\partial x^2} + \frac{2RT\kappa^{\text{eff}}(t^+ - 1)}{F} \left(1 + \frac{d \ln f_{\pm}}{d \ln c_e} \right) \frac{\partial^2 \ln c_e}{\partial x^2} + j_f = 0$ $j_f = a_s i_0 \left(\exp \left(\frac{\alpha_a F}{RT} \eta \right) - \exp \left(-\frac{\alpha_c F}{RT} \eta \right) \right), \quad i_0 = F k c_e^{\alpha_a} (c_{s,\max} - c_{s,e})^{\alpha_a} c_{s,e}^{\alpha_c}$ Separator $\varepsilon_e \frac{\partial c_e}{\partial t} = D_e^{\text{eff}} \frac{\partial^2 c_e}{\partial x^2}$ $\kappa^{\text{eff}} \frac{\partial \varphi_e}{\partial x} + \frac{2RT\kappa^{\text{eff}}(t^+ - 1)}{F} \left(1 + \frac{d \ln f_{\pm}}{d \ln c_e} \right) \frac{\partial \ln c_e}{\partial x} = \frac{I}{A}$	$D_s \frac{\partial c_s}{\partial r} \Big _{r=0} = 0, D_s \frac{\partial c_s}{\partial r} \Big _{r=R_s} = -\frac{j_f}{a_s F} \quad (1)$ $\frac{\partial c_e}{\partial x} \Big _{x=0} = \frac{\partial c_e}{\partial x} \Big _{x=L} = 0 \quad (2)$ $-\sigma^{\text{eff}} \frac{\partial \varphi_s}{\partial x} \Big _{x=0} = \sigma^{\text{eff}} \frac{\partial \varphi_s}{\partial x} \Big _{x=L} = \frac{I}{A}, \frac{\partial \varphi_s}{\partial x} \Big _{x=L_n+L_s} = \frac{\partial \varphi_s}{\partial x} \Big _{x=L_n+L_s} = 0 \quad (3)$ $\frac{\partial \varphi_e}{\partial x} \Big _{x=0} = \frac{\partial \varphi_e}{\partial x} \Big _{x=L} = 0 \quad (4)$ $c_e \Big _{x=L_n^-} = c_e \Big _{x=L_n^+}, D_{e,n}^{\text{eff}} \frac{\partial c_e}{\partial x} \Big _{x=L_n^-} = D_{e,s}^{\text{eff}} \frac{\partial c_e}{\partial x} \Big _{x=L_n^+}, \quad (6)$ $c_e \Big _{x=L_n+L_s^-} = c_e \Big _{x=L_n+L_s^+}, D_{e,s}^{\text{eff}} \frac{\partial c_e}{\partial x} \Big _{x=L_n+L_s^-} = D_{e,p}^{\text{eff}} \frac{\partial c_e}{\partial x} \Big _{x=L_n+L_s^+}$ $\varphi_e \Big _{x=L_n^-} = \varphi_e \Big _{x=L_n^+}, \kappa_{e,n}^{\text{eff}} \frac{\partial \varphi_e}{\partial x} \Big _{x=L_n^-} = \kappa_{e,s}^{\text{eff}} \frac{\partial \varphi_e}{\partial x} \Big _{x=L_n^+}, \quad (7)$ $\varphi_e \Big _{x=L_n+L_s^-} = \varphi_e \Big _{x=L_n+L_s^+}, \kappa_{e,s}^{\text{eff}} \frac{\partial \varphi_e}{\partial x} \Big _{x=L_n+L_s^-} = \kappa_{e,p}^{\text{eff}} \frac{\partial \varphi_e}{\partial x} \Big _{x=L_n+L_s^+}$

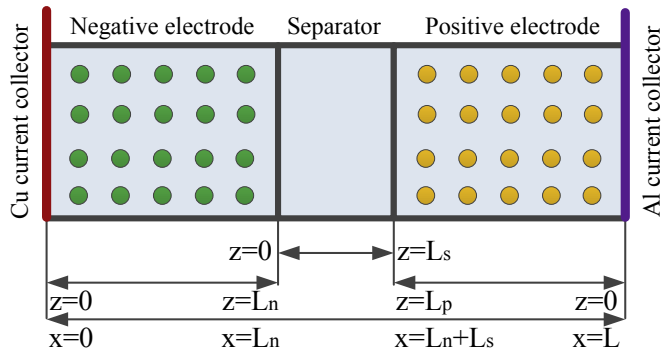


Fig. 1. Schematic diagram to describe the orientation of x and z in each region.

2. Pseudo-two-dimensional model

The P2D porous-electrode model is a rigorous physics-based model, which consists of four coupled partial differential equations and one algebraic equation. These four coupled PDEs together govern the conservation of lithium and conservation of charge in both solid and electrolyte phases, while the Butler-Volmer algebraic equation describes the electrochemical kinetic in the reaction interface [6,7]. As the coupled PDEs in P2D model drastically increase the computing burden of running the model, it has to be simplified to be integrated into an embedded BMS. Table 1 concludes the governing equations of the P2D model for lithium-ion battery.

In the P2D model, the main dimension is the cross-sectional distance x across negative electrode, separator and positive electrode; the additional pseudo dimension is the radial distance r into a specific electrode particle. In order to make the derived formulas more general and also convenient to calculation, the variable z is defined in electrode and separator, respectively. Fig. 1 illustrates orientation of x and z variables in each region. It is noted that the

direction of z in positive electrode is inverse to its direction in other two regions, which makes the derived formulas for negative electrode also suitable to the positive electrode after multiplying the load current I by -1 .

The lithium-ion battery with graphite anode (Gr) and $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (NCM) cathode is examined in this paper, which has been widely applied in electric vehicles. A complete set of model parameters of a 1.78 A h Gr/NCM [30] cell given in Table 2 are employed in this research. The electrolyte dynamic related parameters refer to the literature [31], and are highly dependent on the electrolyte concentration and temperature.

3. Electrolyte concentration approximation

The distribution of electrolyte concentration is first approximated by the parabolic functions, which will pave the way for the latter calculations of reaction flux and electrolyte potential. The parabolic approximation method is developed and validated by Han et al. [26], which can calculate the coefficients of polynomial functions without presupposing the uniform reaction flux. According to the simulation results of the rigorous P2D model shown in section 6, the distribution of electrolyte concentration along x -axis imitates the parabolic profiles. In this section, the concentration dependent variable x is replaced by z , which will simplify the calculation of reaction flux in next section, and make the derived formula for negative electrode also suitable for positive electrode. The expressions of electrolyte concentration for three regions can be assumed as,

$$\begin{cases} c_{e,n}(z) = a_2 z^2 + a_1 z + a_0, 0 \leq z \leq L_n \\ c_{e,s}(z) = a_5 z^2 + a_4 z + a_3, 0 \leq z \leq L_s \\ c_{e,p}(z) = a_8 z^2 + a_7 z + a_6, 0 \leq z \leq L_p \end{cases} \quad (8)$$

Then according to the boundary conditions in Eq. (2), the coefficients of a_1 and a_7 must equal zero, thus Eq. (8) can be simplified as,

Table 2
A set of model parameters for the P2D model.

Symbol (Unit)	Negative	Separator	Positive
$L(\text{m})$	40×10^{-6}	25×10^{-6}	36.55×10^{-6}
$A(\text{m}^2)$	0.102		
$R_s(\text{m})$	5×10^{-6}		5×10^{-6}
ε_s	0.662		0.58
ε_e	0.3	0.4	0.3
$c_{s,\max}(\text{mol m}^{-3})$	3.108×10^4		5.183×10^4
θ_0	0.001		0.9554
θ_{100}	0.7908		0.3597
$c_{e,0}(\text{mol m}^{-3})$	1.2×10^3		
$k(\text{mol}^{-0.5}\text{m}^{2.5}\text{s}^{-1})$	2×10^{-11}		2×10^{-11}
α_a, α_c	0.5, 0.5		0.5, 0.5
t^+	0.38		
$D_s(\text{m}^2\text{s}^{-1})$	1.4×10^{-14}		2×10^{-14}
$\sigma(\text{Sm}^{-1})$	100		100
$R_{SEI}(\Omega\text{m}^2)$	0		0
$d \ln f_{\pm} / d \ln c_e$	1.5732		
$D_e(\text{m}^2\text{s}^{-1}) = 10 \left(-8.43 - \frac{54}{T - 5 \times 10^{-3} c_e - 229} - 0.22 \times 10^{-3} c_e \right)$			
$\kappa(\text{Sm}^{-1}) = c_e \left[\frac{(-10.5 + 0.074T - 6.96 \times 10^{-5}T^2)}{c_e^2(4.94 \times 10^{-7} - 886 \times 10^{-12}T^2)} + c_e(668 \times 10^{-6} - 17.8 \times 10^{-6}T - 0.028 \times 10^{-6}T^2) + \right]^2 \times 10^{-7}$			
$U_p(V) = -10.72\theta^4 + 23.88\theta^3 - 16.77\theta^2 + 2.595\theta + 4.563$ for $\theta \in (0, 1)$			
$U_n(V) = 0.1493 + 0.8493 \exp(-61.79\theta) + 0.3824 \exp(-665.8\theta) - \exp(39.42\theta - 41.92)$			
$-0.03131 \tan^{-1}(25.59\theta - 4.099) - 0.009434 \tan^{-1}(32.49\theta - 15.74)$ for $\theta \in (0.3, 1)$			

$$\begin{cases} c_{e,n}(z) = a_1 z^2 + a_0, 0 \leq z \leq L_n \\ c_{e,s}(z) = a_4 z^2 + a_3 z + a_2, 0 \leq z \leq L_s \\ c_{e,p}(z) = a_6 z^2 + a_5, 0 \leq z \leq L_p \end{cases} \quad (9)$$

where $[a_0, a_1, a_2, a_3, a_4, a_5, a_6]$ are the time-varying coefficients need to be calculated for each sampling period. According to the boundary conditions in Eq. (6), four linear equations can be constructed as,

$$a_1 L_n^2 + a_0 = a_2 \quad (10)$$

$$a_6 L_p^2 + a_5 = a_4 L_s^2 + a_3 L_s + a_2 \quad (11)$$

$$2a_1 L_n D_{e,n}^{\text{eff}} = a_3 D_{e,s}^{\text{eff}} \quad (12)$$

$$-2a_6 L_p D_{e,p}^{\text{eff}} = (2a_4 L_s + a_3) D_{e,s}^{\text{eff}} \quad (13)$$

where $D_{e,n}^{\text{eff}}$ is the effective diffusion coefficient with Bruggeman relation $D_{e,n}^{\text{eff}} = D_{e,n} \varepsilon_n^{1.5}$ to account for the tortuous effect by porous electrode and separator, and the negative sign in Eq. (13) considers the inverse direction of derivative. By integrating the functions of electrolyte concentration along the z -axis, the total amount of the Li^+ in the three regions can be acquired as,

$$Q_{e,n}(t) = \varepsilon_{e,n} \int_0^{L_n} c_{e,n}(z) dz = \varepsilon_{e,n} \left(\frac{1}{3} a_1 L_n^3 + a_0 L_n \right) \quad (14)$$

$$Q_{e,s}(t) = \varepsilon_{e,s} \int_0^{L_s} c_{e,s}(z) dz = \varepsilon_{e,s} \left(\frac{1}{3} a_4 L_s^3 + \frac{1}{2} a_3 L_s^2 + a_2 L_s \right) \quad (15)$$

$$Q_{e,p}(t) = \varepsilon_{e,p} \int_0^{L_p} c_{e,p}(z) dz = \varepsilon_{e,p} \left(\frac{1}{3} a_6 L_p^3 + a_5 L_p \right) \quad (16)$$

Before solving the linear Eqs. (10)–(16), $Q_{e,n}(t)$, $Q_{e,s}(t)$, $Q_{e,p}(t)$ need to be obtained for each sampling period. Firstly, taking use of the charge conservation and integrating Eq. (3) along the whole electrode for both sides, following equations can be obtained,

$$\int_0^{L_n} j_f(z) dz = \frac{I}{A} \quad (17)$$

$$\int_0^{L_p} j_f(z) dz = -\frac{I}{A} \quad (18)$$

where j_f is the local volumetric current density and equal to the reaction flux j multiple with specific interfacial surface area ($a_s = 3\varepsilon_s/R_s$) and Faraday constant (F), namely $j_f = a_s F j$. Then by integrating Eq. (2) and Eq. (6) along the whole electrode and separator regions, and taking Eqs. (17) and (18) into, the total amount of the Li^+ can be calculated as,

$$\frac{d}{dt} Q_{e,n}(t) = (1 - t_+) \frac{I}{AF} + 2a_1 L_n D_{e,n}^{\text{eff}} \quad (19)$$

$$\frac{d}{dt} Q_{e,s}(t) = 2a_4 L_s^2 D_{e,s}^{\text{eff}} \quad (20)$$

$$\frac{d}{dt} Q_{e,p}(t) = -(1 - t_+) \frac{I}{AF} + 2a_6 L_p^2 D_{e,p}^{\text{eff}} \quad (21)$$

Since the initial distribution of electrolyte concentration is known and keeps uniform along the three regions. Thus the initial total amount of the Li^+ satisfy,

$$Q_{e,n}(0) = \varepsilon_{e,n} c_{e,0} L_n, Q_{e,s}(0) = \varepsilon_{e,s} c_{e,0} L_s, Q_{e,p}(0) = \varepsilon_{e,p} c_{e,0} L_p \quad (22)$$

And the initial coefficients of Eq. (9) satisfy,

$$[a_0, a_1, a_2, a_3, a_4, a_5, a_6] = [c_{e,0}, 0, c_{e,0}, 0, 0, c_{e,0}, 0] \quad (23)$$

By applying the Forward Euler method to solve Eqs. (19)–(21), the $Q_{e,n}(t)$, $Q_{e,s}(t)$, $Q_{e,p}(t)$ at next sampling period can be obtained, then the coefficients of electrolyte concentration functions can be calculated by solving the linear Eqs. (10)–(16). Repeating the above process, the distribution of electrolyte concentration at any time can be obtained.

4. Reaction flux approximation

As the battery terminal voltage consists of four parts, and all of them are determined by the reaction flux at the two current collector interfaces. In lower current rates, the reaction flux can be assumed as uniform distribution, while as the increase of current rates, the non-uniformity of reaction flux become prominent [28,29]. The above reality manifest that the non-uniform distribution of reaction flux needs to be considered to maintain high fidelity of battery model especially at higher current rates, which often happen in driving cycles. In this section, the polynomial approximation technology is employed to obtain the non-uniform distribution of reaction flux, which presents the superiority of lower computing burden. The following contents illustrate the proposed method in detail, and for the concise expression, the variable j_f is used rather than j .

4.1. Preparation knowledge

In each electrode, the activation overpotential at any position can be described from the potential balance equation,

$$\eta(z) = \varphi_s(z) - \varphi_e(z) - U(z) - \frac{R_{SEI}}{a_s} j_f(z) \quad (24)$$

Also according to Eq. (5), the activation overpotential can be derived as the form of inverse hyperbolic sine,

$$\eta(z) = \frac{2RT}{F} \text{asinh} \left(\frac{j_f(z)}{2a_s i_0} \right) \quad (25)$$

According to Eqs. (24) and (25), the overpotential can be eliminated; then by taking the derivative for both sides, the following relationship can be constructed,

$$P(z) \frac{\partial j_f(z)}{\partial z} = \frac{\partial \varphi_s(z)}{\partial z} - \frac{\partial \varphi_e(z)}{\partial z} - \frac{\partial U(z)}{\partial z} \quad (26)$$

where

$$P(z) = \left(\frac{RT}{2a_s Fi_0} \frac{1}{\sqrt{1 + \left(\frac{j_f(z)}{2a_{s10}} \right)^2}} + \frac{R_{SEI}}{a_s} \right) \quad (27)$$

and can be calculated by taking into the previous estimation of j_f . By integrating Eqs. (3) and (4) along the z -axis, the derivative of solid and electrolyte phase potential can be acquired as,

$$\frac{\partial \varphi_s(z)}{\partial z} = \frac{1}{\sigma^{eff}} \int_0^z j_f dz - \frac{I}{\sigma^{eff} A} \quad (28)$$

$$\frac{\partial \varphi_e(z)}{\partial z} = -\frac{2RT(t_+ - 1)}{F} \frac{\partial \ln c_e(z)}{\partial z} - \frac{1}{\kappa^{eff}} \int_0^z j_f dz \quad (29)$$

where σ^{eff} is the effective solid phase conductivity with Bruggeman relation $\sigma^{eff} = \sigma_s^{1.5}$, and κ^{eff} is the effective electrolyte phase conductivity with Bruggeman relation $\kappa^{eff} = \kappa_e^{1.5}$. According to Table 2, the equilibrium potential of each electrode can be calculated as,

$$U(z) = U(\theta(z)) = U\left(\frac{c_{s,e}(z)}{c_{s,max}}\right) \quad (30)$$

where $c_{s,e}$ is the solid phase surface concentration, which can be calculated by using the historical values of j_f , and the detailed method will be discussed in next section. Then by applying the finite difference theory, the derivative of equilibrium potential at different z points can be calculated as,

$$\begin{aligned} \frac{\partial U(z)}{\partial z} \Big|_{z=0} &= \frac{4U(\Delta z) - 3U(0) - U(2\Delta z)}{2\Delta z}, \\ \frac{\partial U(z)}{\partial z} \Big|_{z=\frac{L_n}{2}} &= \frac{U\left(\frac{L_n}{2} + \Delta z\right) - U\left(\frac{L_n}{2} - \Delta z\right)}{2\Delta z}, \\ \frac{\partial U(z)}{\partial z} \Big|_{z=L_n} &= \frac{U(L_n - 2\Delta z) - 4U(L_n - \Delta z) + 3U(L_n)}{2\Delta z} \end{aligned} \quad (31)$$

where the accuracy of the derivative at boundary is improved by replacing the 2nd-order term in Taylor series expression with the 2nd-order central difference.

According to the simulation results of the rigorous P2D model shown in section 6, the distribution of j_f along x -axis imitates the parabolic or cubic profile, and the similar results are also obtained in the literature of [28]. So in next subsections, the 2nd-order and 3rd-order polynomial approximation methods are proposed, and the accuracy of them will be compared in section 6. In the process of deriving undetermined coefficients for polynomial functions, the negative electrode is only considered. For the positive electrode, simply multiply the current I by -1 , and replace the negative electrode parameters in the equations correspondingly.

4.2. Parabolic approximation

For 2nd-order polynomial approximation, the j_f can be assumed as,

$$j_f(z) = c_2 z^2 + c_1 z + c_0 \quad (32)$$

To determine coefficients of Eq. (32), take Eqs. (28) and (29) into Eq. (26), set z to equal zero and L_n respectively, and take use of the boundary conditions of Eqs. (3) and (4), then two linear equations can be acquired as,

$$P(0)c_1 = -\frac{I}{\sigma^{eff} A} - \frac{\partial U(z)}{\partial z} \Big|_{z=0} \quad (33)$$

$$P(L_n)(2c_2 L_n + c_1) = \frac{2RT}{F} \frac{\partial \ln c_e(z)}{\partial z} \Big|_{z=L_n} + \frac{I}{\kappa^{eff} A} - \frac{\partial U(z)}{\partial z} \Big|_{z=L_n} \quad (34)$$

where,

$$\frac{\partial \ln c_e(z)}{\partial z} = \frac{2a_1 z}{a_1 z^2 + a_0} \quad (35)$$

according to Eq. (9). Besides, according to Eq. (17), the last one equation for calculating the coefficients can be derived as,

$$\frac{c_2 L_n^3}{3} + \frac{c_1 L_n^2}{2} + c_0 L_n = \frac{I}{A} \quad (36)$$

By solving the linear Eqs. (33), (34) and (36), the distribution of local volumetric current density can be acquired.

4.3. Cubic approximation

For 3rd-order polynomial approximation, the j_f can be assumed as,

$$j_f(z) = c_3 z^3 + c_2 z^2 + c_1 z + c_0 \quad (37)$$

Likewise, take Eqs. (28) and (29) into Eq. (26), set z to equal zero, $L_n/2$ and L_n respectively, and take use of the boundary conditions of Eqs. (3) and (4), then three linear equations can be acquired as,

$$P(0)c_1 = -\frac{I}{\sigma^{eff} A} - \frac{\partial U(z)}{\partial z} \Big|_{z=0} \quad (38)$$

$$\begin{aligned} P\left(\frac{L_n}{2}\right) \left(3c_3 \frac{L_n^2}{4} + c_2 L_n + c_1 \right) &= \left(\frac{1}{\sigma^{eff}} + \frac{1}{\kappa^{eff}} \right) \int_0^{L_n/2} j_f dz - \frac{I}{\sigma^{eff} A} \\ &\quad + \frac{2RT}{F} \frac{\partial \ln c_e(z)}{\partial z} \Big|_{z=\frac{L_n}{2}} - \frac{\partial U(z)}{\partial z} \Big|_{z=\frac{L_n}{2}} \end{aligned} \quad (39)$$

$$\begin{aligned} P(L_n) \left(3c_3 L_n^2 + 2c_2 L_n + c_1 \right) &= \frac{2RT}{F} \frac{\partial \ln c_e(z)}{\partial z} \Big|_{z=L_n} + \frac{I}{\kappa^{eff} A} \\ &\quad - \frac{\partial U(z)}{\partial z} \Big|_{z=L_n} \end{aligned} \quad (40)$$

where

$$\int_0^z j_f(z) dz = \frac{c_3}{4} z^4 + \frac{c_2}{3} z^3 + \frac{c_1}{2} z^2 + c_0 z \quad (41)$$

Besides, according to Eq. (18), the last one equation for calculating the coefficients can be derived as,

$$\frac{c_3 L_n^4}{4} + \frac{c_2 L_n^3}{3} + \frac{c_1 L_n^2}{2} + c_0 L_n = \frac{I}{A} \quad (42)$$

Finally, by solving the linear Eqs. (38)–(40), (42), the distribution of local volumetric current density can be acquired.

5. Polynomial approximation model

In previous sections, the approximation methods of electrolyte

concentration and the non-uniform of reaction are illustrated. In this section, the methods of calculating solid phase surface concentration, activation overpotential, electrolyte potential and terminal voltage are elaborated, thus a polynomial approximation pseudo-two-dimensional model (PP2D) with high fidelity and lower computing burden is constructed.

5.1. Solid phase surface concentration

In the literature, lots of methods have been proposed to obtain the solid phase surface concentration with different computational burden and fidelity. In this paper, the PP method introduced by Subramanian [17] is selected to calculate the solid phase surface concentration due to its better trade-off between approximation accuracy and computing complexity. The formulas of the three-parameter PP model are concluded as,

$$\frac{d}{dt}\bar{c}_s(t) + 3\frac{j_f}{a_sFR_s} = 0 \quad (43)$$

$$\frac{d}{dt}\bar{q}(t) + 30\frac{D_s}{R_s^2}\bar{q}(t) + \frac{45}{2}\frac{j_f}{a_sFR_s^2} = 0 \quad (44)$$

$$35\frac{D_s}{R_s}[c_{s,e}(t) - \bar{c}_s(t)] - 8D_s\bar{q}(t) + \frac{j_f}{a_sF} = 0 \quad (45)$$

where $c_{s,e}(t)$ is the surface concentration, $\bar{c}_s(t)$ is the volume-averaged concentration and $\bar{q}(t)$ is the volume-averaged concentration flux. The initial $\bar{c}_s(t)$ can be calculated according to the initial SOC,

$$\bar{c}_s(0) = (\theta_0 + SOC_0(\theta_{100} - \theta_0))c_{s,max} \quad (46)$$

and the initial $\bar{q}(t)$ can be set as zero.

5.2. Activation overpotential

After obtaining the local volumetric current density, the activation overpotential at different x points can be calculated according to Eq. (25).

5.3. Electrolyte potential

Take use of Eq. (7) and Eq. (29), and integrate the equations along the whole regions for both sides, the electrolyte potential can be derived as,

$$\varphi_e(z) - \varphi_e(0) = \frac{2RT(1-t_+)}{F} \ln \frac{c_e(z)}{c_e(0)} - \frac{1}{\kappa_{eff}} \int_0^z \int_0^z j_f(\xi) d\xi dz, \text{negative/positive} \quad (47)$$

$$\varphi_e(z) - \varphi_e(0) = \frac{2RT(1-t_+)}{F} \ln \frac{c_e(z)}{c_e(0)} - \frac{Iz}{Ak_{eff}}, \text{separator} \quad (48)$$

where

$$\int_0^z \int_0^z j_f(\xi) d\xi dz = \frac{c_3}{20}z^5 + \frac{c_2}{12}z^4 + \frac{c_1}{6}z^3 + \frac{c_0}{2}z^2 \quad (49)$$

According to Eqs. (47) and (48), the electrolyte potential difference between the positive current collector and negative current

collector is obtained as,

$$\varphi_{e,p}(0) - \varphi_{e,n}(0) = \frac{2RT(1-t_+)}{F} \ln \frac{c_{e,p}(0)}{c_{e,n}(0)} + \frac{1}{\kappa_p^{eff}} \int_0^{L_p} \int_0^z j_{f,p}(\xi) d\xi dz - \frac{1}{\kappa_n^{eff}} \int_0^{L_n} \int_0^z j_{f,n}(\xi) d\xi dz - \frac{IL_s}{Ak_s^{eff}} \quad (50)$$

5.4. Terminal voltage

The terminal voltage of battery cell is defined as,

$$V(t) = \varphi_{s,p}(0) - \varphi_{s,n}(0) \quad (51)$$

Take Eq. (24) into Eq. (51), the terminal voltage can be finally expressed as,

$$V(t) = U_p \left(\frac{c_{s,e}^p(0)}{c_{s,max}^p} \right) - U_n \left(\frac{c_{s,e}^n(0)}{c_{s,max}^n} \right) + \eta_p(0) - \eta_n(0) + \varphi_{e,p}(0) - \varphi_{e,n}(0) + \frac{R_{SEI}^p}{a_p} j_{f,p}(0) - \frac{R_{SEI}^n}{a_n} j_{f,n}(0) \quad (52)$$

As expressed in Eq. (52), the terminal voltage of battery consists of four parts: the equilibrium potential, the activation overpotential, the electrolyte potential, and the overpotential caused by the SEI film resistance. It is noted that the accuracy of these four parts can be improved by using the non-uniform reaction flux, which has been calculated in section 4.

A calculation flowchart of the proposed model is presented in Fig. 2 to clarify the battery simulation process, where Δt is the sampling step and t_{max} is the maximum simulation time. At the beginning time, the battery is assumed in the steady-state, so the electrolyte concentration, electrode concentration and local volumetric current density are uniform; the stoichiometry of electrode concentration can be derived according to the initial SOC. Taking use of these initial conditions, c_e at next sampling step is calculated based on Eqs. (9)–(16), then j_f at next sampling step is obtained by the proposed parabolic approximation or cubic approximation method. After that, η and $\Delta\phi_e$ can be derived directly based on Eq. (25) and Eq. (50) respectively, $c_{s,e}$ can be calculated according to the PP method illustrated in section 5.1. Then U of each electrode can be obtained based on Eq. (30), and the derivative of U at different z points are employed to calculate j_f at next sampling step. By aggregating the above variables, terminal voltage can be acquired based on Eq. (52). Repeating the above calculation process, the simulation terminates with the time up to the maximum simulation time.

6. Results and discussion

In this work, the original P2D model is simulated by the commercial software COMSOL Multiphysics 5.1a, which solves the PDEs based on finite element method. The accuracy of COMSOL has been verified by many researchers [13,14,24,26], and its results can be looked as the benchmark for comparison analysis. The SP model and proposed PP2D model can be easily created and subsequently simulated in MATLAB, which is a powerful tool for modeling, control and simulation. An ordinary personal computer (Intel Core i5-3210 M CPU 2.5 GHz with 2.71 GB RAM available) is used to

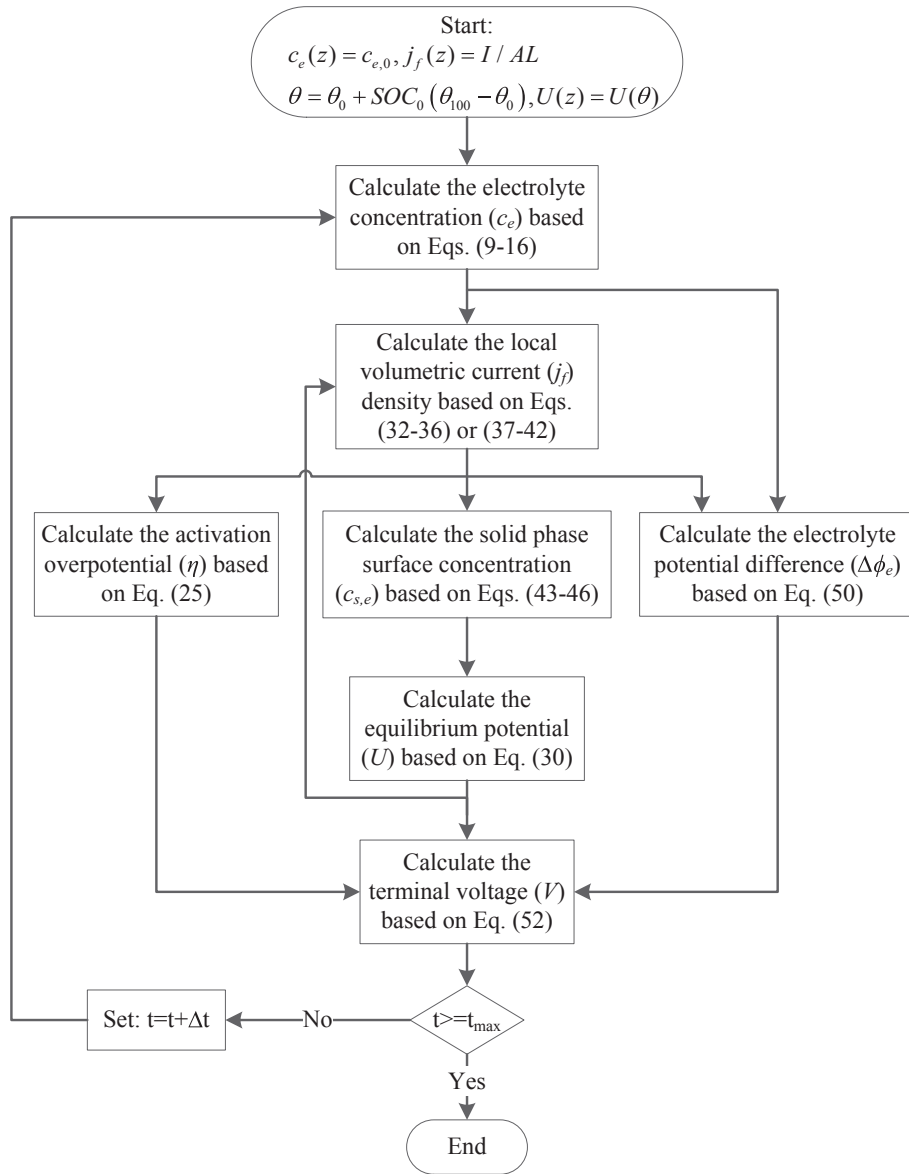


Fig. 2. The calculation flowchart of the proposed model.

simulate these models with the sampling period set as 1 s. Besides the constant current operations, the hybrid pulse and UDDS cycles are employed to incite the battery models. The UDDS cycle represents city driving conditions for light duty vehicles and has been widely used to evaluate the battery models. Fig. 3 illustrates the

used current profiles, where the profile of UDDS cycle is extracted from a commercial electric vehicle, and then scaled down to adapt to the battery. To evaluate performance of the proposed model, an aggressive hybrid pulse reaching a maximum of 10 C-rates is applied. The UDDS and constant current cycles are simulated with

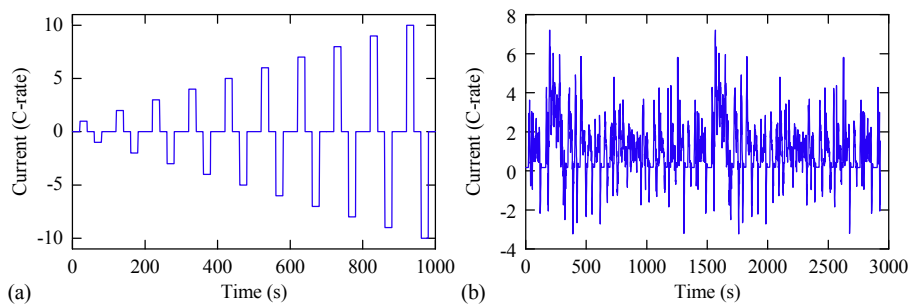


Fig. 3. The two typical current profiles. (a) Hybrid pulse cycle; (b) UDDS cycle.

initial SOC equal to 100%, while the hybrid pulse begins with 50% SOC to avoid exceeding the cut-off voltage.

6.1. Validation of electrolyte concentration

The electrolyte concentration simulated by approximation method and P2D model is illustrated in Fig. 4. To evaluate the approximation ability of parabolic functions, Fig. 4(a) presents the concentration distribution along the x -axis under 1C discharge at different moments, while Fig. 4(b) shows the corresponding concentration variation with respect to time at four interfaces of battery. Under 1C discharge, the electrolyte concentration of the studied battery reaches the steady state after 20 s, and the parabolic functions provide acceptable estimation of the electrolyte concentration distribution at different moments. As the time goes by, the concentration variation at four interfaces is efficiently traced with negligible error.

The electrolyte concentration simulated at hybrid pulse and UDDS cycles is presented in Fig. 4(c) and (d), the larger amplitude of current in pulse and frequently changing current in UDDS cycle imitate the strict driving conditions in vehicle. Since the electrolyte potential difference between the negative and positive current collectors is determined by the corresponding electrolyte concentration, the concentration variation with respect to time at these two interfaces is examined. Compared with the rigorous model, satisfactory results are obtained by the approximation method both for hybrid pulse and UDDS cycles. Especially, the estimation concentrations keep much close to the results from rigorous model at strictly dynamic load. It has to be admitted that the concentration deviation exists especially at heavy current than 5 C-rates, which might be caused due to the only use of the electrolyte diffusion equation and the lack of restriction from the electrolyte potential equation. Although not perfect, the above results verify the approximation method is meaningful and worthy of being integrated into a BMS.

6.2. Validation of non-uniform current

The distribution of local volumetric current density along x -axis

at 2C discharge is simulated for the rigorous and approximation models. The results at four different discharge moments are illustrated in Fig. 5. To evaluate the approximation ability clearly, the deviation values relative to the results of SP model are plotted in Fig. 5. It can be seen that the current density at different position along the x -axis is non-uniform and the distribution profile also changes with time. At the initial time in Fig. 5(a), the current density follows the parabolic distribution, and both the 2nd-order (2nd app) and 3rd-order approximation (3rd app) agree well with the results from P2D model. However, at the time $t = 500$ s, the current density at negative electrode behaves the cubic distribution. The 2nd app cannot imitate this profile, and thus make larger error; while the 3rd app still can capture this profile with less error. It is noted that both at $t = 1000$ s and $t = 1500$ s in negative electrode, the errors of 3rd app are larger than that of 2nd app. This phenomenon might be explained as when the current density follows parabolic distribution, the 2nd app is adequate to capture the profile; while the additional parameter in 3rd app brings some computational errors, which in turn decreases the estimation accuracy of current density.

The local volumetric current density at negative current collector (cc-neg) and positive current collector (cc-pos) are shown in Fig. 6, while the uniform current density in SP model is also plotted in the figure for comparison. As illustrated in Fig. 6, both the 2nd app and 3rd app can imitate the profile of current density at two current collector interfaces with some errors. It can be seen the accuracy of 3rd app is better than the accuracy of 2nd app in cc-neg; while the accuracy of them are comparable in cc-pos. And as the increase of discharge rate, the deviation between the rigorous model and approximation method also increases. Although owning non-negligible errors, the proposed approximation methods successfully predict the variation trend of non-uniform current and present obvious superiority over the SP model. Due to the more focus on current density at current collectors, thus the 3rd app presents more strength than 2nd app.

6.3. Validation of potential and voltage

The simulation results of activation overpotential difference

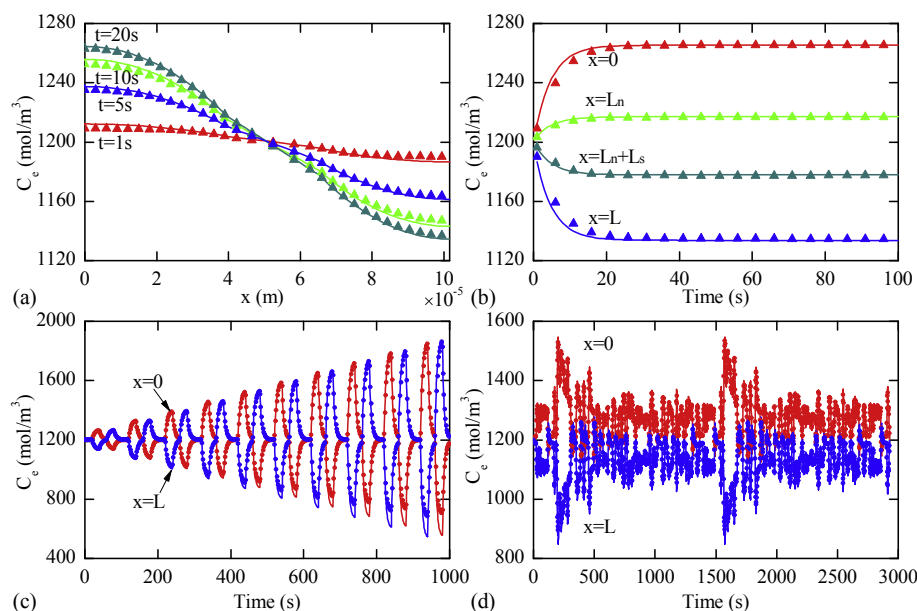


Fig. 4. Comparison of the electrolyte concentration simulated by approximation method (solid line) and P2D (marker). (a) Concentration distributions along x -axis; (b) concentration variation with respect to time; (c) hybrid pulse cycle; (d) UDDS cycle.

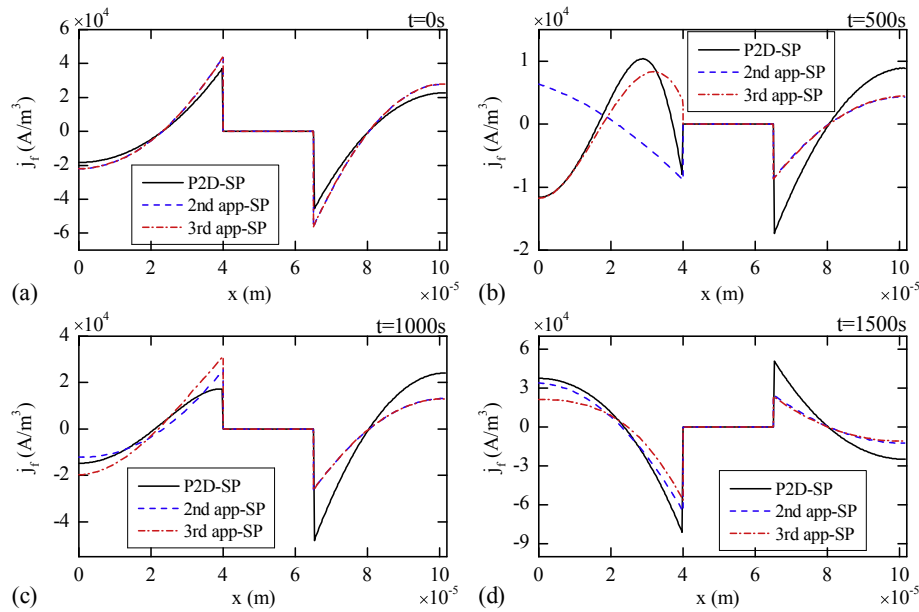


Fig. 5. Comparison of the distribution of j_f along x -axis at 2C discharge. (a) $t = 0$ s; (b) $t = 500$ s; (c) $t = 1000$ s; (d) $t = 1500$ s.

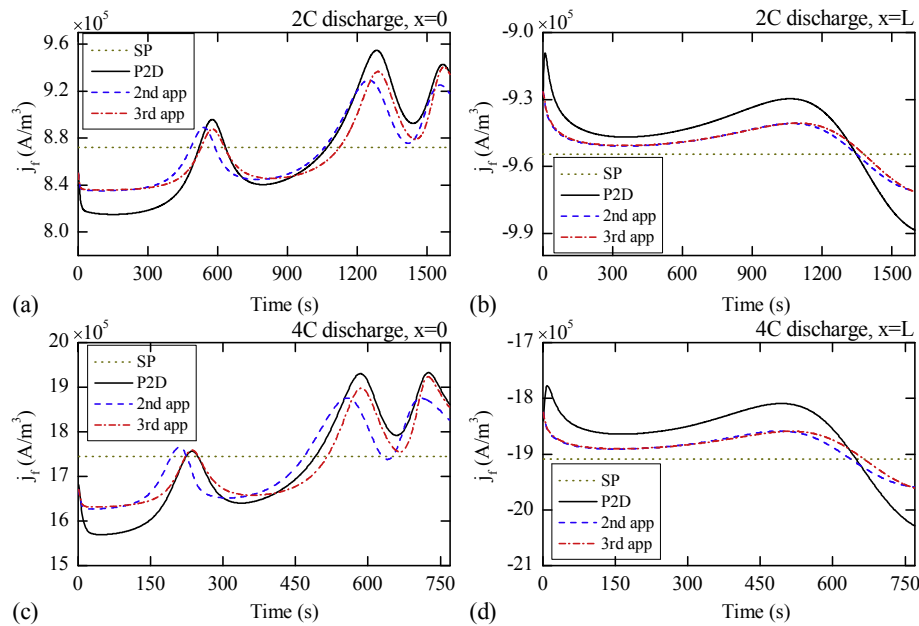


Fig. 6. Comparison of j_f at cc-neg and cc-pos interfaces. (a) cc-neg at 2C discharge; (b) cc-pos at 2C discharge; (c) cc-neg at 4C discharge; (d) cc-pos at 4C discharge.

between two current collector interfaces for hybrid pulse and UDDS cycles are illustrated in Fig. 7, and for the convenience of comparison, the deviation values of P2D and PP2D relative to the results of SP model are plotted in the figure. It can be seen that the actual overpotential from rigorous model exists much variation due to the non-uniform current density verified in Fig. 5. The results show that the overpotential errors caused by assuming uniform current density in SP model are not insignificant. As illustrated in Fig. 7 (a) and its zoom graph Fig. 7 (b), the deviation beyond 5 mV at 2C discharge/charge condition, which is unacceptable for a strict BMS. Also it can be observed that the proposed PP2D model could compensate over half of the deviation both in hybrid pulse and UDDS cycles.

The simulation results of electrolyte potential difference between two current collector interfaces for hybrid pulse and UDDS cycles are illustrated in Fig. 8. It is noted that in SP model, the electrolyte potential is assumed as zero. For hybrid pulse cycle shown in Fig. 8(a) and (b), the results of PP2D model agree well with the results from the rigorous model. The electrolyte potential difference is over 10 mV for 1C pulse, which is too much to be ignored for a BMS. This is the main reason why the SP model only adapts to the operation lower than 1C rates. For PP2D model, the errors are negligible for pulse lower than 6C discharge/charge, while obvious error exists for pulse beyond 6C discharge/charge. The main reason is that the approximation of electrolyte concentration produces relatively higher error under heavy current as

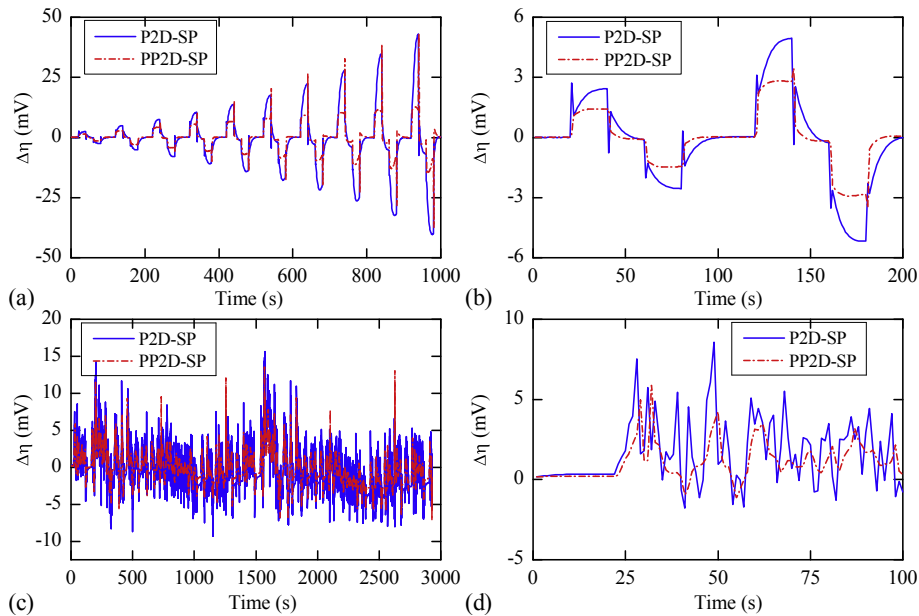


Fig. 7. Comparison of the activation overpotential difference between two current collector interfaces for two typical cycles. (a) Hybrid pulse cycle; (b) zoo picture of pulse cycle; (c) UDDS cycle; (d) zoo picture of UDDS cycle.

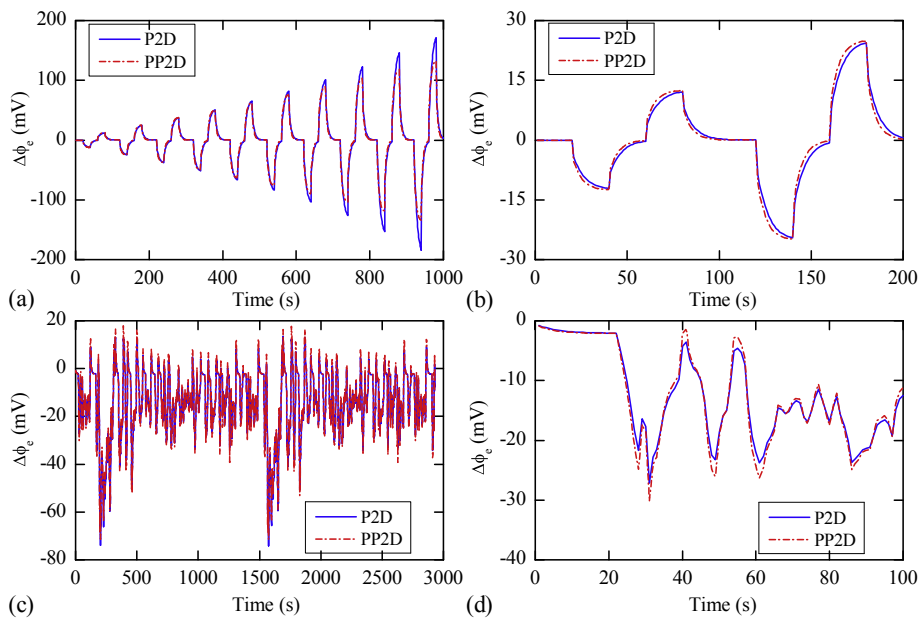


Fig. 8. Comparison of the electrolyte potential difference between two current collector interfaces for two typical cycles. (a) Hybrid pulse cycle; (b) zoo picture of pulse cycle; (c) UDDS cycle; (d) zoo picture of UDDS cycle.

presented in Fig. 4 (c). For UDDS cycle shown in Fig. 8(c) and (d), the PP2D model can also capture the electrolyte potential variation accurately even under the frequently changing load.

The simulation results of terminal voltages are plotted in Fig. 9, and the numeric results of voltage errors are listed in Table 3. It is noted that the terminal voltages embody the combined effect of equilibrium potential, the electrolyte potential, the activation overpotential, and the SEI film overpotential on the external characteristic of battery. As illustrated in pulse cycle in Fig. 9(a), obvious errors exist in SP model when compared with results of P2D model. Additionally, 1C pulse causes above 10 mV voltage error for SP model in Fig. 9(b), which validates the fact that the original SP

model only keep accurate at rates lower than 1C. For pulse cycle, the maximum error (MAE) of SP model beyond 140 mV, while the MAE of PP2D model only reaches 32 mV; the root mean square error (RMSE) of SP model over 42 mV, while the RMSE of PP2D model just presents 7.3 mV. The results verify the proposed PP2D model can imitate the rigorous P2D model efficiently in a broad current range. As for the results of UDDS cycle, the superiority of PP2D model over SP model is much obvious, and the MAE and RMSE of PP2D model only reaches 20.9 mV and 3.2 mV, respectively. The results verify the proposed PP2D model can successfully trace the rigorous P2D model under the strictly dynamic load.

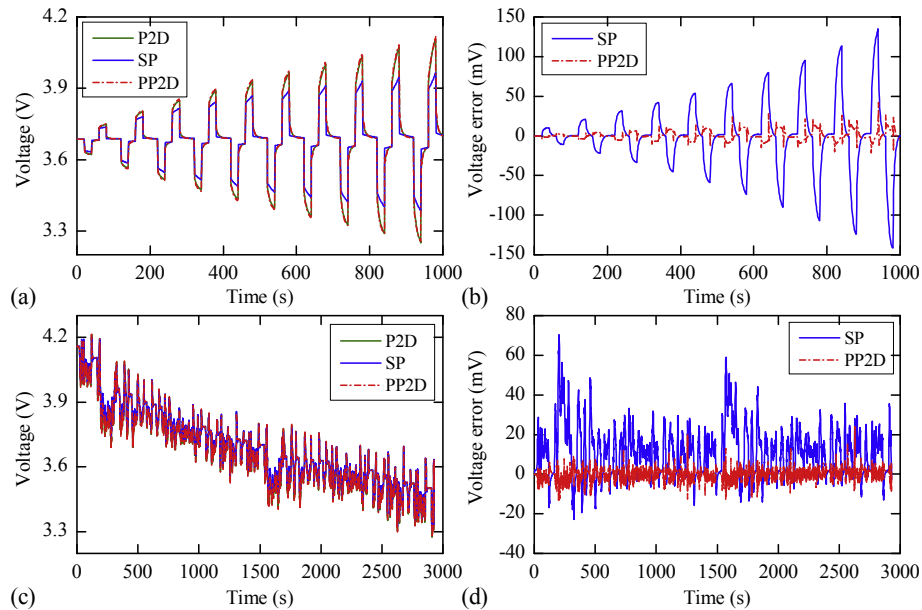


Fig. 9. Comparison of the terminal voltage simulated with P2D and PP2D models for two typical cycles. (a) Hybrid pulse cycle; (b) voltage error of pulse cycle; (c) UDDS cycle; (d) voltage error of UDDS cycle.

Table 3
Comparison of the terminal voltage errors for the SP and PP2D models.

	MaxE (mV)		RMSE (mV)	
	SP	PP2D	SP	PP2D
Hybrid pulse cycle	141.4	32.0	42.1	7.3
UDDS cycle	70.4	20.9	16.9	3.2

6.4. Validation of computing efficiency

In the PP2D model, the PDEs are approximated by a series of polynomial functions, which make it to be a lumped-parameter model. Thus arithmetic operations with small computation complexity can be used to simulate the model. To evaluate the improvement of computing efficiency, the simulation times of SP, P2D and PP2D models under different operation conditions are compared in Table 4. Under the constant current conditions, the simulation is conducted from 100% SOC to 10% SOC, thus the computation times at lower C-rates are more than that at higher C-rates. As listed in Table 4, the computation times of PP2D model are only about one sixtieth of the P2D model and lower than twice of SP model. The simulation step is set as 1 s, and the calculation time of PP2D model for every step is within 2 ms, which make it feasible to be applied into an embedded BMS. Compared with the constant current conditions, more times are needed to simulate the dynamic cycle for the P2D model, which not happen for SP and PP2D model. It can be understood that the numerical method (finite element method in COMSOL) to solve the P2D model needs plenty of

repeated calculation to keep the convergence for dynamic profile inputs; while for lumped-parameter models, no repeated calculation are needed. In general, the above results verify that the proposed PP2D model significantly improves the computing efficiency of running the battery model.

7. Conclusion

In the field of battery system engineering, physics-based modes have been expected to implement in the near future due to their ability to give insight into internal electrochemical states of battery. Acquiring a model with high fidelity, lower computational complexity and facing entire operation range is crucial for current researches. In this paper, a series of polynomial functions are employed to simplify the rigorous P2D model of lithium-ion battery.

The electrolyte concentration is first approximated by parabolic functions without assuming the uniform current density, which paves the way for the latter calculations of reaction flux and electrolyte potential. The simulation results verify that the satisfactory results are acquired with negligible error for both hybrid pulse and UDDS cycle. The distribution of reaction flux (evaluated by local volumetric current density) is approximated by 2nd-order and 3rd-order polynomial functions, and the polynomial coefficients are obtained by solving the derived linear equations without any other assumption. The results validate that the proposed approximation methods successfully predict the variation trend of non-uniform current, thus presents obvious superiority over the SP model. However, non-negligible errors exist for the proposed method. In

Table 4
Comparison of the simulation times for the SP, P2D and PP2D models.

Operation conditions (duration time)	SP	P2D	PP2D	Ratio (P2D/PP2D)	Ratio (PP2D/SP)
1C discharge (3400s)	2.93s	365s	5.23s	69.79	1.78
2C discharge (1600s)	1.62s	180s	2.52s	71.43	1.56
4C discharge (770s)	0.82s	78s	1.31s	59.54	1.60
Hybrid pulse cycle (1000s)	1.06s	181s	1.63s	111.04	1.54
UDDS cycle (2930s)	2.72s	785s	4.33s	181.29	1.59

further study, other mathematic functions such as exponential function and trigonometric function will be attempted to improve the approximation accuracy.

Benefit from the acquisition of electrolyte concentration and non-uniform reaction flux, the electrolyte potential is derived directly according to the conservation of charge at electrolyte phase; the accuracy of activation overpotential is improved when compared with the SP model using uniform reaction flux. The proposed PP2D is compared with the rigorous P2D model and original SP model under different constant current rates, hybrid pulse and UDDS cycles. Under 10C discharge/charge pulse, the MAE of terminal voltage for SP model beyond 140 mV, while only 32 mV for PP2D model. At UDDS cycle, the RMSE of terminal voltage for SP model up to 16.9 mV, while just about 3.2 mV for PP2D model. Meanwhile, the computation time of PP2D model is only one sixtieth of the P2D model, lower than twice of SP model and within 2 ms for every simulation step. These results demonstrate that the model proposed in this paper imitates the rigorous P2D model efficiently and can be integrated into an embedded BMS.

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